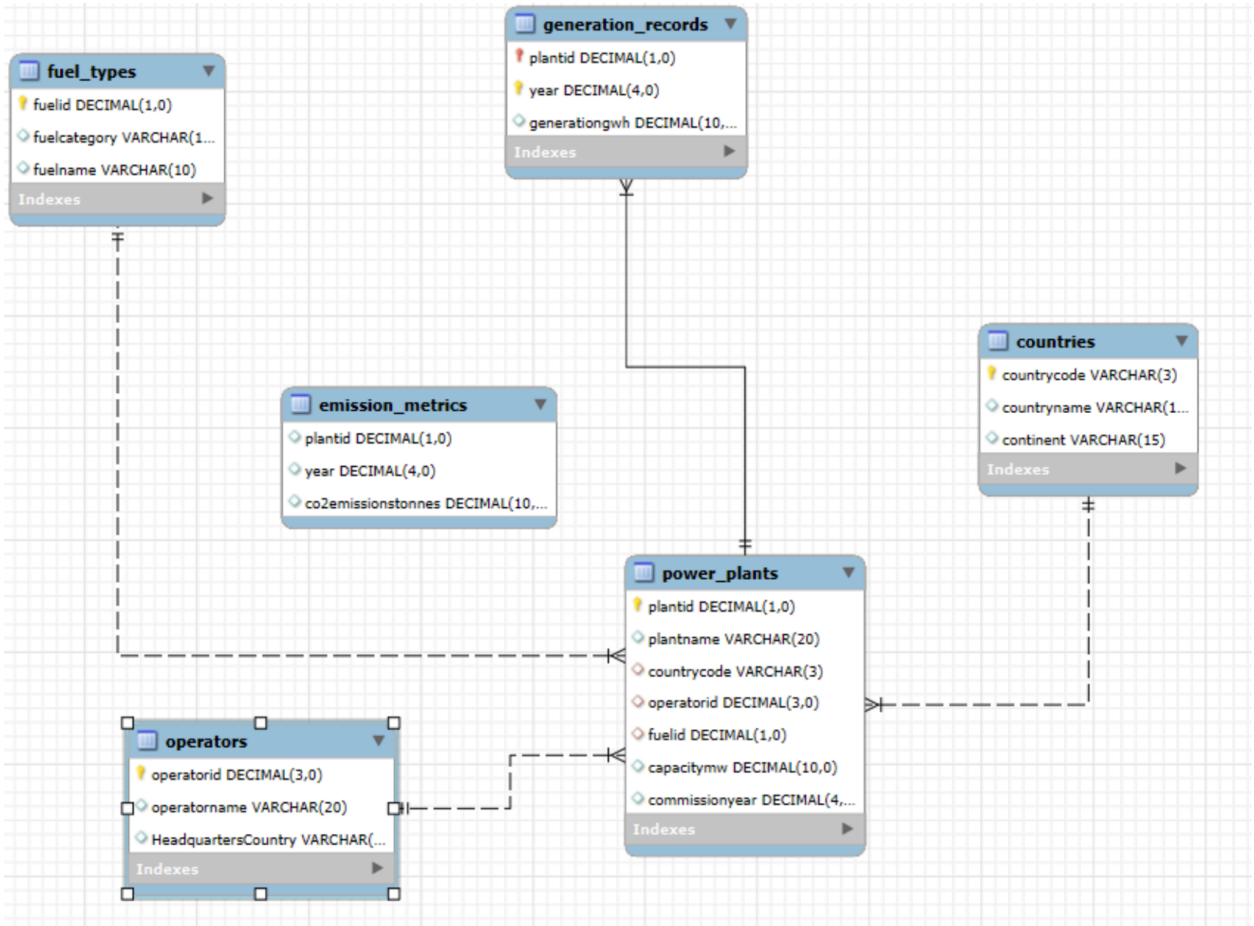
Hailey Newbold

Assignment 3

EER Diagram



1

```sql
-- 1
select power_plants.plantname, countries.countryname, operators.operatorname, fuel_types.fuelcategory,
       fuel_types.fuelname, power_plants.capacitymw, power_plants.commissionyear
from power_plants
    inner join countries on power_plants.countrycode = countries.countrycode
    inner join operators on power_plants.operatorid = operators.operatorid
    inner join fuel_types on power_plants.fuelid = fuel_types.fuelid
order by power_plants.capacitymw desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

plantname	countryname	operatorname	fuelcategory	fuelname	capacitymw	commissionyear
Three Gorges	China	China Yangtze Power	Renewable	Hydro	22500	2003
Itaipu	Brazil	Electrobras	Renewable	Hydro	14000	1984
Tuoketuo	China	China Yangtze Power	Fossil	Coal	13200	1995
Kashiwazaki-Kariwa	Japan	Tepco Energy	Nuclear	Uranium	8212	1985
Grand Coulee	United States	Global Hydro Inc	Renewable	Hydro	6809	1942
Vindhyachal	India	NTPC Limited	Fossil	Coal	4760	1987
Janschwalde	Germany	Rhein Power AG	Fossil	Coal	3000	1981
Loy Yang	Australia	Snowy Hydro	Fossil	Coal	2210	1984

2

```sql
-- 2
select power_plants.plantname, power_plants.countrycode,
       generation_records.year, generation_records.generationgwh
from power_plants
    inner join generation_records on power_plants.plantid = generation_records.plantid
order by generation_records.generationgwh desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

plantname	countrycode	year	generationgwh
Three Gorges	CHN	2024	98800.00
Itaipu	BRA	2024	89000.10
Tuoketuo	CHN	2024	71000.30
Vindhyachal	IND	2024	35600.80
Grand Coulee	USA	2024	21500.50
Janschwalde	DEU	2024	18400.20
Loy Yang	AUS	2024	14200.40
Kashiwazaki-Kariwa	JPN	2024	0.00

3

```sql
-- 3
select power_plants.plantid, power_plants.countrycode, generation_records.year,
       generation_records.year, emission_metrics.co2emissionstonnes
from power_plants
    inner join generation_records on power_plants.plantid = generation_records.plantid
    inner join emission_metrics on power_plants.plantid = emission_metrics.plantid
order by emission_metrics.co2emissionstonnes asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

plantid	countrycode	year	year	co2emissionstonnes
1	USA	2024	2024	0.00
4	BRA	2024	2024	0.00
5	JPN	2024	2024	0.00
7	CHN	2024	2024	0.00
6	AUS	2024	2024	1950000.00
2	DEU	2024	2024	2450000.00
3	IND	2024	2024	3100000.00
8	CHN	2024	2024	5200000.00

4

```sql
-- 4
with operatorgeneration as (
    select operators.operatorname, operators.headquarterscountry,
           sum(generation_records.generationgwh) as total
    from power_plants
        inner join operators on power_plants.operatorid = operators.operatorid
        inner join generation_records on power_plants.plantid = generation_records.plantid
    group by operators.operatorname, operators.headquarterscountry
)
select OperatorName, headquarterscountry, total
from operatorgeneration
order by total desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

OperatorName	headquarterscountry	total
China Yangtze Power	CHN	169800.30
Electrobras	BRA	89000.10
NTPC Limited	IND	35600.80
Global Hydro Inc	USA	21500.50
Rhein Power AG	DEU	18400.20
Snowy Hydro	AUS	14200.40
Tepco Energy	JPN	0.00